



Swiss Institute of
Bioinformatics

ECCB Elixir track

Bottom up interoperability using semantics in ELIXIR resources

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Why





AS/NZS 3112

NEMA 5-15

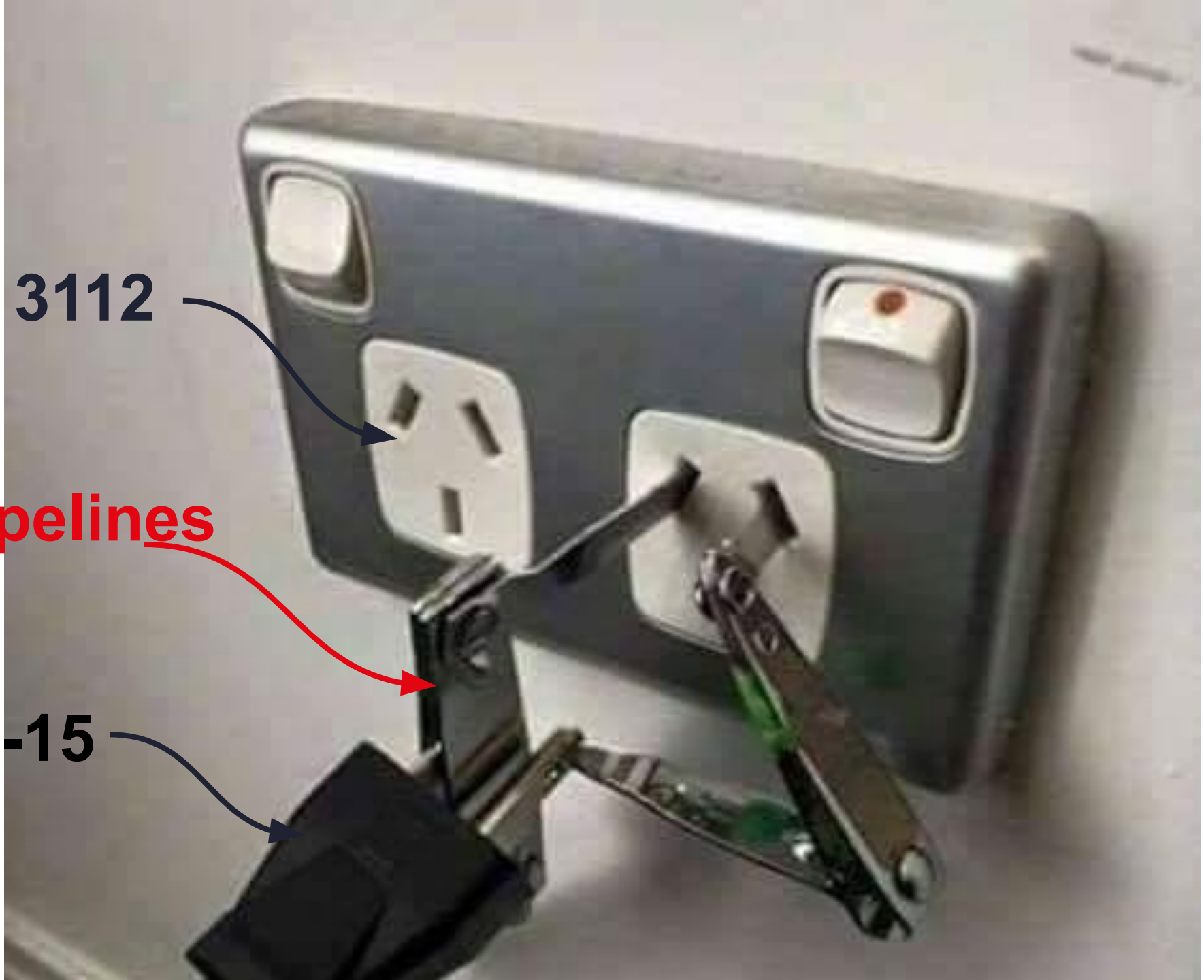




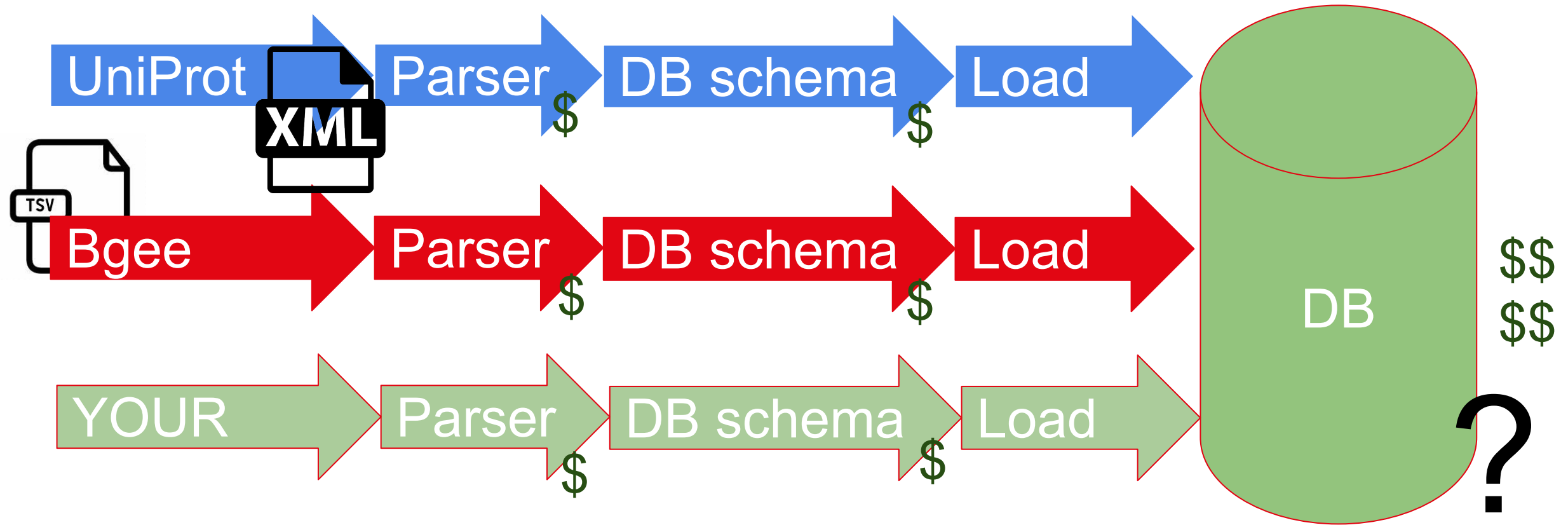
AS/NZS 3112

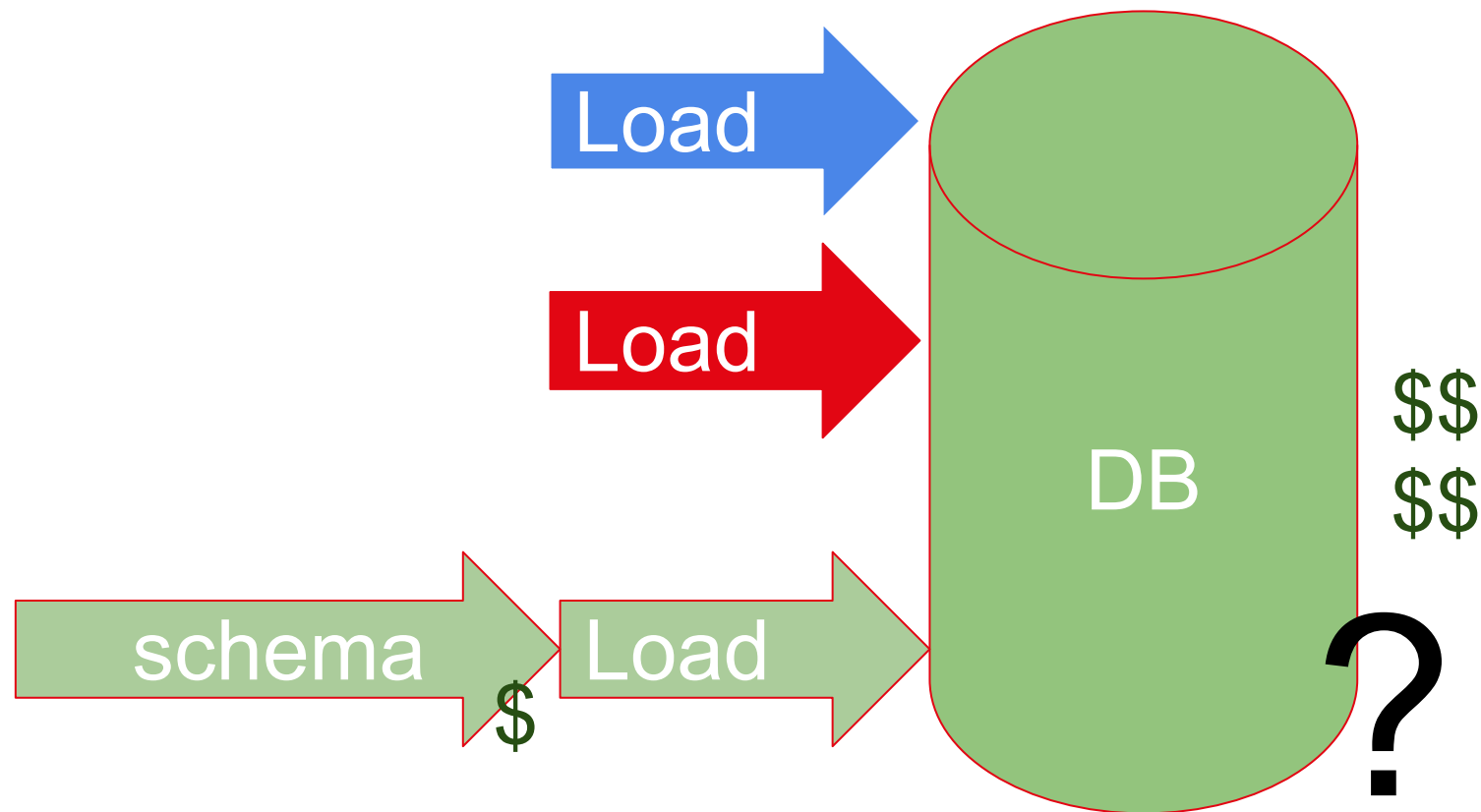
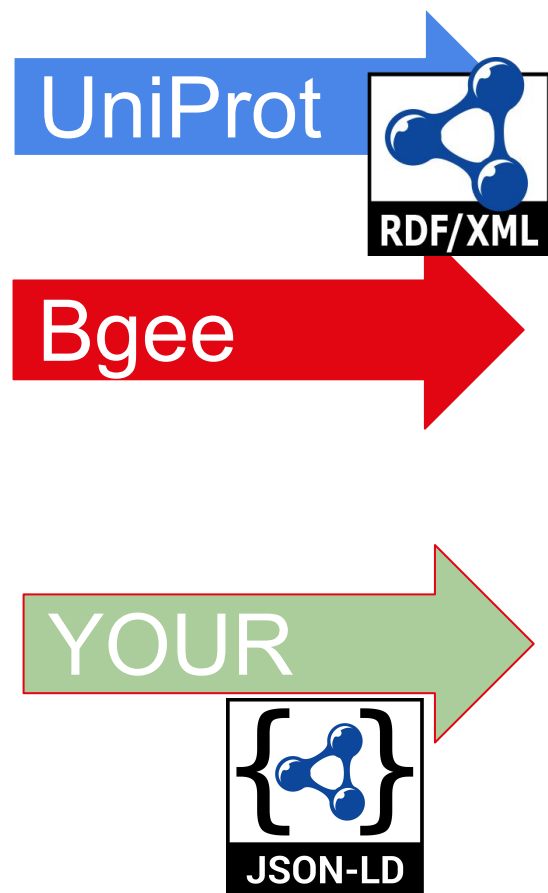
ETL pipelines

NEMA 5-15

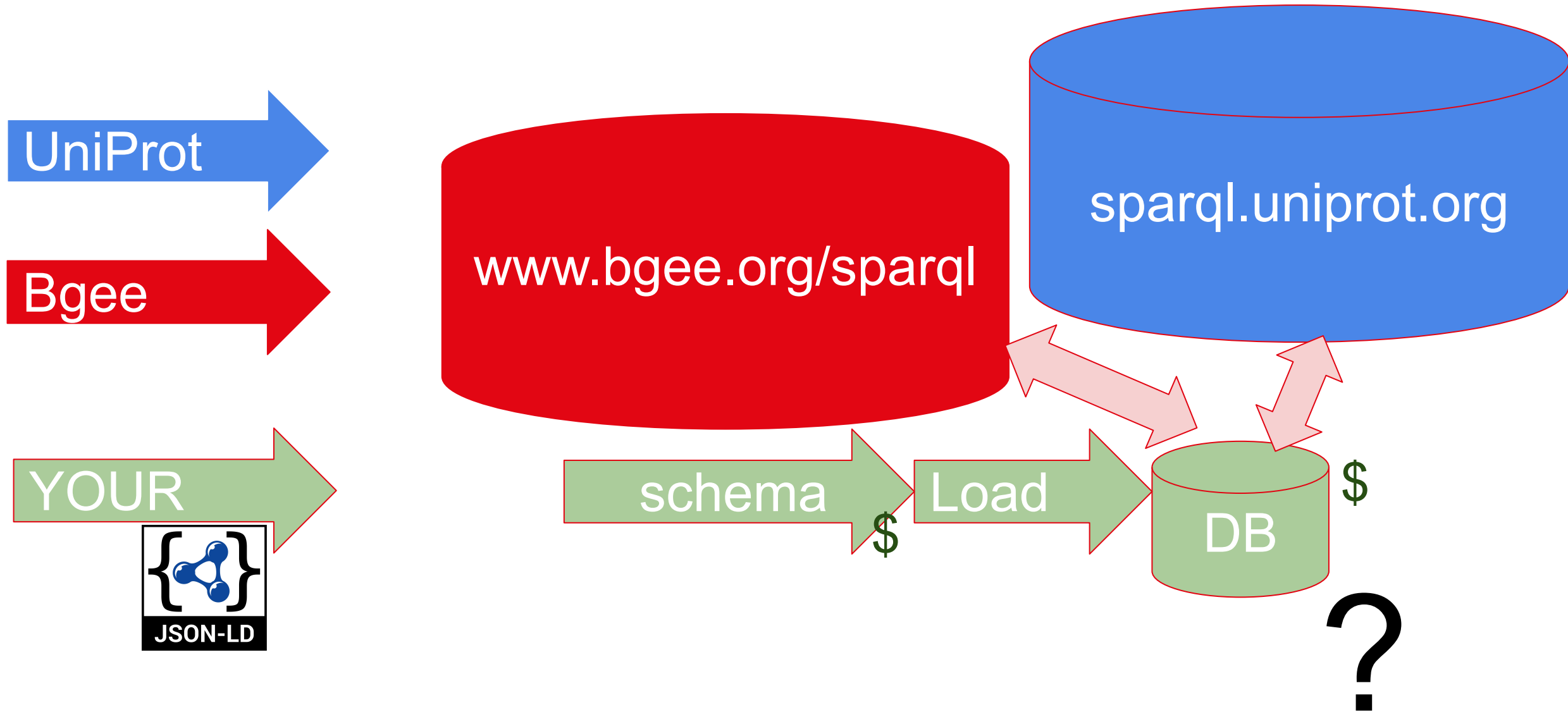


 Parse, parse, parse





RDF + Federated SPARQL





RDF is so common you don't even see it

Who



CDR, Interoperability, deposition, Services with RDF

 OMA Browser

  UniProt

 Cellosaurus

 ChEMBL

 LipidMaps

 BRENDA

 BacDive

 Rhea

 ChEBI

 Bgee

 InterMine

 Identifiers.org

 DisProt

 SIBiLS

  EDAM

 HAMAP

 HumanMine

 MolMeDB

 IDSM / Schem

 SwissLipids

 WikiPathways.org

 Data Catalogue

 OrthoDB

 AgroLD

Subset of each category that are findable as announced services by nodes



Hidden RDF/Semantics

- (Bio)Schema.org
- PDF
- Flight details and hotel bookings in e-mails
- Spreadsheets and office documents
- Photos and images
- Ontologies
- Grant data

FAIR:11

What

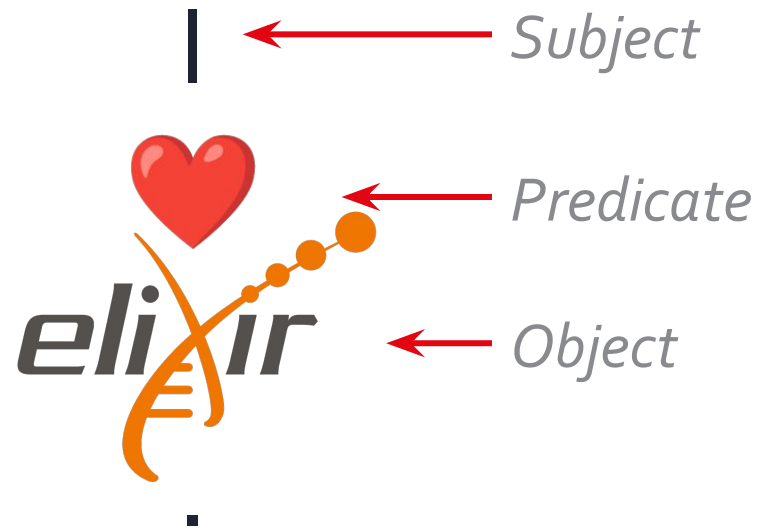


Basic RDF a sentence





Basic RDF a sentence





Basic RDF use identifiers for concepts

<<https://orcid.org/0000-0002-7449-1266>>

<<http://www.wikidata.org/entity/Q316>>

<<https://ror.org/o44rwnt51>>

.

FAIR: F1



Declare your abbreviations

prefix orcid: <<https://orcid.org/>>

prefix wd: <<http://www.wikidata.org/entity/>>

prefix ror: <<https://ror.org/>>

[orcid:0000-0002-7449-1266](https://orcid.org/0000-0002-7449-1266)

[wd:Q316](http://www.wikidata.org/entity/Q316)

[ror:044rwnt51](https://ror.org/044rwnt51) .



love (Q316)



Full identifier gives access to more data

Item [Discussion](#)



strong, positive emotion based on affection
affection

▼ In more languages

[Configure](#)

Language	Label	Description	Also known as
default for all languages	No label defined	–	
English	love	strong, positive emotion based on affection	affection
American English	love	strong, positive emotion based on affection	affection

FAIR: A2

Abuse your abbreviation power!

prefix jerven:<<https://orcid.org/0000-0002-7449-1266>>

prefix wd:<<http://www.wikidata.org/entity/Q316>>

prefix ELIXIR:<<https://ror.org/o44rwnt51>>

jerven:



ELIXIR:

.



Semantics via Rules or Logic

prefix owl:<<http://www.w3.org/2002/07/owl#>>

prefix hate:<<http://www.wikidata.org/entity/L4473>>

wd:[Q316](http://www.wikidata.org/entity/Q316):

owl:[disjointWith](#)

[hate](#):

.



Why now?



ExpasyGPT **Beta**

ExpasyGPT is an AI-driven tool that facilitates queries across life science databases in Expasy. It also provides accurate information on bioinformatics resources developed by SIB. This allows you to easily:

- identify resources for your use case
- compile biological information from different databases*
- ask complex questions of one database*

The tool either answers your question directly, or converts it into a SPARQL query that you can then automatically run across relevant databases on Expasy and/or further edit.

* See [how ExpasyGPT works](#), including which databases it can query.

ExpasyGPT is in Beta version and may make mistakes or experience issues. We are continually improving its performance, and **highly value your input!** Please send your suggestions or questions to the [Expasy helpdesk](#).

Ask your question



Which SIB resources are supported by ExpasyGPT?

Where is the ACE2 gene expressed in humans?

List primate genes expressed in the fruit fly eye

What are the rat orthologs of the human TP53 gene?

What is the HGNC symbol for the P68871 protein?

Anatomical entities where the INS zebrafish gene is expressed and their gene GO annotations



Example / Use case

Find **disease** related **enzymes**
catalyzing **reactions**, the
metabolites and the potential **drugs**
acting on them



UniProt for Disease (genetic linked to OMIM)

SELECT

?protein ?disease ?rhea ?metabolite ?drug

WHERE {

?protein up:annotation **?diseaseAnnotation** , ?catalyticActivityAnnotation .

?diseaseAnnotation a up:Disease Annotation ; up:disease **?disease** .

?catalyticActivityAnnotation up:catalyticActivity/up:catalyzedReaction ?rhea .

SERVICE <https://sparql.rhea-db.org/sparql/> {

?rhea rh:side/rh:contains/rh:compound ?compound .

?compound (rh:chebi|(rh:reactivePart/rh:chebi)|(rh:underlyingChebi/rh:chebi))

?metabolite .

}

SERVICE <https://idsm.elixir-czech.cz/sparql/endpoint/idsm> {

[] a cco:Activity ; cco:hasMolecule ?drug ;

cco:hasAssay/cco:hasTarget/cco:hasTargetComponent/cco:targetCmptXref ?protein

.

}

}



UniProt to Rhea (reactions: balanced and in vivo)

SELECT

?protein ?disease **?rhea** ?metabolite ?drug

WHERE {

?protein up:annotation ?diseaseAnnotation , **?catalyticActivityAnnotation** .

?diseaseAnnotation a up:Disease_Annotation ; up:disease ?disease .

?catalyticActivityAnnotation up:catalyticActivity/up:catalyzedReaction **?rhea** .

SERVICE <https://sparql.rhea-db.org/sparql/> {

?rhea rh:side/rh:contains/rh:compound ?compound .

?compound (rh:chebi|(rh:reactivePart/rh:chebi)|(rh:underlyingChebi/rh:chebi))

?metabolite .

}

SERVICE <https://idsm.elixir-czech.cz/sparql/endpoint/idsm> {

[] a cco:Activity ; cco:hasMolecule ?drug ;

cco:hasAssay/cco:hasTarget/cco:hasTargetComponent/cco:targetCmptXref ?protein

.

}

}

UniProt to Metabolite (ChEBI via Rhea)

SELECT

?protein ?disease **?rhea** **?metabolite** ?drug

WHERE {

?protein up:annotation ?diseaseAnnotation , **?catalyticActivityAnnotation** .

?diseaseAnnotation a up:Disease_Annotation ; up:disease ?disease .

?catalyticActivityAnnotation up:catalyticActivity/up:catalyzedReaction **?rhea** .

SERVICE <https://sparql.rhea-db.org/sparql/> {

?rhea rh:side/rh:contains/rh:compound **?compound** .

?compound (rh:chebi | (rh:reactivePart/rh:chebi) | (rh:underlyingChebi/rh:chebi))

?metabolite .

}

SERVICE <https://idsm.elixir-czech.cz/sparql/endpoint/idsm> {

[] a cco:Activity ; cco:hasMolecule ?drug ;

cco:hasAssay/cco:hasTarget/cco:hasTargetComponent/cco:targetCmptXref ?protein

.

}

}

UniProt to drug (ChEMBL via IDSM)

SELECT

?protein ?disease ?rhea ?metabolite **?drug**

WHERE {

?protein up:annotation ?diseaseAnnotation , ?catalyticActivityAnnotation .
?diseaseAnnotation a up:Disease_Annotation ; up:disease ?disease .
?catalyticActivityAnnotation up:catalyticActivity/up:catalyzedReaction ?rhea .

SERVICE <https://sparql.rhea-db.org/sparql/> {

?rhea rh:side/rh:contains/rh:compound ?compound .

?compound (rh:chebi|(rh:reactivePart/rh:chebi)|(rh:underlyingChebi/rh:chebi))

?metabolite .

}

SERVICE <https://idsm.elixir-czech.cz/sparql/endpoint/idsm> {

[] a cco:Activity ; cco:hasMolecule **?drug** ;

cco:hasAssay/cco:hasTarget/cco:hasTargetComponent/cco:targetCmptXref **?protein**

.

}

}

UniProt + Rhea + ChEBI + ChEMBL

SELECT

?protein **?disease** **?rhea** **?metabolite** **?drug**

WHERE {

?protein up:annotation **?diseaseAnnotation** , **?catalyticActivityAnnotation** .

?diseaseAnnotation a up:Disease_Annotation ; up:disease **?disease** .

?catalyticActivityAnnotation up:catalyticActivity/up:catalyzedReaction **?rhea** .

SERVICE <https://sparql.rhea-db.org/sparql/> {

?rhea rh:side/rh:contains/rh:compound **?compound** .

?compound (rh:chebi | (rh:reactivePart/rh:chebi) | (rh:underlyingChebi/rh:chebi))

?metabolite .

}

SERVICE <https://idsm.elixir-czech.cz/sparql/endpoint/idsm> {

[] a cco:Activity ; cco:hasMolecule **?drug** ;

cco:hasAssay/cco:hasTarget/cco:hasTargetComponent/cco:targetCmptXref **?protein**

.

}

}

SELECT

?protein **?disease** **?rhea** **?metabolite** **?drug**

WHERE {

?protein up:annotation **?diseaseAnnotation** , **?catalyticActivityAnnotation** .

?diseaseAnnotation a up:Disease_Annotation ; up:disease **?disease** .

?catalyticActivityAnnotation up:catalyticActivity/up:catalyzedReaction **?rhea** .

SERVICE <https://sparql.rhea-db.org/sparql/> {

?rhea rh:side/rh:contains/rh:compound **?compound** .

?compound (rh:chebi | (rh:reactivePart/rh:chebi) | (rh:underlyingChebi/rh:chebi))

?metabolite .

}



SERVICE <https://idsm.elixir-czech.cz/sparql/endpoint/idsm> {

[] a cco:Activity ; cco:hasMolecule **?drug** ;

cco:hasAssay/cco:hasTarget/cco:hasTargetComponent/cco:targetCmptXre

.

}

}





protein	disease	rhea	metabolite	drug
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5303356 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5199380 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5087479 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5070897 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL4559961 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5170047 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5085070 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5396270 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL4297611 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	http://rdf.ebi.ac.uk/resource/chembl/molecule/CHEMBL5396270 
http://purl.uniprot.org/uniprot/Q9NVM4 	http://purl.uniprot.org/diseases/4865 	http://rdf.rhea-db.org/48100 	http://purl.obolibrary.org/obo/CHEBI_59789 	

Drug

protein
disease
reaction
metabolite
drug
... connected

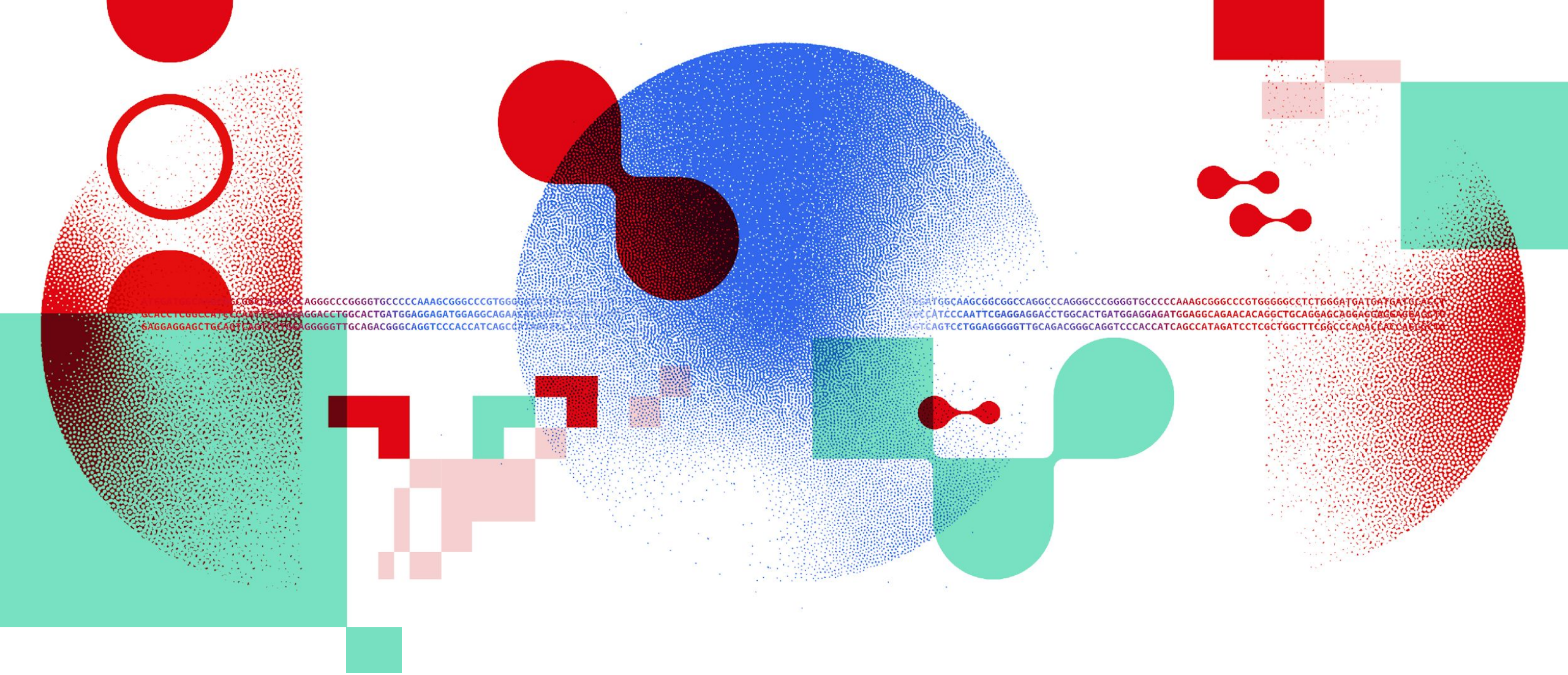
More examples →





Don't give up on complex queries:

If a complex search query fails, try to use SPARQL instead of giving up.



Thank you & all ELIXIR members

DATA SCIENTISTS FOR LIFE

sib.swiss